Bot Hunter Analysis Report

1. Executive Summary

Bot Hunter analysed 149,239 click events.
It selected 3,731 as likely bot traffic, which is 2.50% of the run.

This report should be read as evidence for review, not as proof of fraud for every individual event. The dataset does not include ground-truth labels, so the probability figures are operational confidence estimates rather than measured precision or recall.

The submitted output is submission.tsv. It keeps the required binary is_bot decision and adds an operational tier so the same prediction can be used more carefully in business workflows.

2. What The System Does

Bot Hunter combines two complementary classifiers.

- Rules-based classifier: transparent rules that identify repeated, mechanically timed, or tightly clustered click patterns.
- Machine-learning classifier: an Extended Isolation Forest anomaly model over 14 engineered behavioural features.

The final score is a weighted blend:

```text
combined_score = (0.58 * heuristic_score) + (0.42 * ml_score)
```

The rules layer is weighted slightly higher because it is easier to explain and audit. The anomaly model still matters because it can identify unusual combinations that a small rule set may not capture.

3. Main Results

The strongest explainable patterns in this run were:

- repeated query: 3,403 events
- repeated query/domain pair: 2,086 events
- confirmed query repetition: 2,023 events

- very short query: 1,798 events
- high-volume clicked domain: 1,732 events
- heavy region/browser/OS cluster: 1,168 events
- concentrated ct context: 904 events
- same-second click burst: 677 events
- moderately long time-to-click: 431 events
- dense burst repetition cluster: 236 events

Example interpretation: a single repeated query is not enough to prove bot traffic. It becomes more concerning when it appears with a repeated clicked domain, dense same-second activity, reused time-to-click values, or a narrow region/browser/OS footprint.

The new concentrated ct rule is deliberately cautious. It adds low-weight support only when country-like concentration appears with repeated query behaviour and either a heavy device cluster or a same-second burst. Country-like concentration alone is not treated as bot evidence because legitimate campaigns can be country-specific.

4. Classifier Details

The rules layer currently scores:

- repeated query/domain pairs
- repeated queries
- confirmed query repetition
- high-volume clicked domains
- dense region/browser/OS clusters
- exact time-to-click reuse
- same-second bursts
- dense burst repetition clusters
- concentrated ct context paired with repetition and clustering
- implausibly fast clicks
- moderately long or extreme time-to-click values
- regular pseudo-session inter-arrival timing

Rule contributions are separated into strong and supporting evidence. Strong rules represent direct mechanical or replay-like patterns, such as impossible timing or repeated query/domain behaviour. Supporting rules provide context, such as volume, concentration, or broad clustering. Supporting contributions are capped so several weak signals cannot add up to the same meaning as a strong bot signal on their own.

Rule strength	Meaning	Score treatment
Strong	Direct mechanical or replay evidence	Contributes its full rule weight
Supporting	Contextual or weaker evidence	Capped together at 0.24

The exact time-to-click reuse rule uses a 99th-percentile reuse-count threshold with an absolute floor. The main repetition and concentration rules also use 99th-percentile batch thresholds with the previous fixed count and total-rate cutoffs kept as guardrails. This lets the rules adapt to the observed population while avoiding weak duplicate counts in small runs.

The ct rule, confirmed query repetition, and dense burst repetition rules are conjunctive. In practical terms, they require multiple signals to be present before adding score. This is safer than treating broad context, such as country-like concentration, as a standalone bot indicator.

The anomaly classifier uses the engineered feature matrix. The main feature families are:

- region/browser/OS frequency
- global ct country frequency
- sub-200 ms click flags
- local burst density
- query entropy
- query, domain, and query/domain repetition counts
- same-second and exact time-to-click reuse counts
- timing magnitude after log transformation
- low-cardinality kp and sld value frequencies

High-volume domain frequency and global country frequency are down-weighted to 0.50 and 0.50, respectively. The low-cardinality kp and sld frequency features are down-weighted to 0.50 and 0.25. Events isolated quickly by random hyperplane splits receive higher anomaly scores. EIF is the only production anomaly model; alternate ML backends and supervised pilots have been removed.

kp and sld are treated as categorical-style assumptions rather than true continuous measurements because their observed cardinality is very low. Raw kp and sld values remain in the feature artefact for audit, but the anomaly model uses log_kp_count and log_sld_count instead of raw numeric distances. The raw values are excluded from ml_feature_names.

Threshold-change validation used the regenerated submission.tsv, artifacts/summary.json, artifacts/features.tsv, artifacts/sample_events.json, and report files under docs. Targeted heuristic, pipeline, and dashboard tests passed (uv run pytest tests/test_heuristics.py tests/test_pipeline.py tests/test_web.py, 42 passed), the full test suite passed (uv run pytest, 48 passed), and Black passed for the touched Python files with the existing Python 3.12 target-version warning. Each generated report reflects the current run's artefacts; fixed before/after comparison metrics are kept in the static README task history rather than repeated in this template.

5. Thresholds And Decision Logic

An event is selected as a bot when either condition is true:

text
combined_score > 0.568053
or heuristic_score >= 0.62

The combined-score threshold is the run-specific 97.5th-percentile cutoff. It keeps the selected volume stable for an unlabelled dataset while letting the ML model influence borderline cases.

The heuristic override protects high-confidence, explainable rule hits. For example, a strongly repeated query/domain pattern with mechanical timing should not be missed only because the anomaly ranking moved after a feature change.

Adaptive heuristic thresholds used in this run:

Rule	Computed threshold	Basis
Concentrated ct context (concentrated_ct_context)	29013	99th-percentile threshold, floor 1000, rate guardrail 14923
Heavy region/browser/OS cluster (heavy_device_cluster)	43674	99th-percentile threshold, floor 600, rate guardrail 5223
High-volume clicked domain (high_volume_domain)	2238	99th-percentile threshold, floor 200, rate guardrail 2238
Repeated query (repeat_query)	149	99th-percentile threshold, floor 12, rate guardrail 149
Repeated query/domain pair (repeat_query_domain)	37	99th-percentile threshold,

floor 4, rate guardrail 37 |
| Reused exact time-to-click (reused_ttc) | 40 | 99th-percentile threshold, floor 40 |
| Same-second click burst (same_second_burst) | 6 | 99th-percentile threshold, floor 4 |

In this run:

- heuristic-only flag rate: 1.36%
- ML agreement-tail reference rate: 2.50%
- operational confidence estimate: 74%

6. Method Agreement And Disagreement

Agreement between the rules layer and the anomaly model is useful review evidence. It is not statistical validation, because there are no labels.

At the ML agreement threshold, this run has 1,726 Heuristic + ML events and 2,005 ML only events.

Method disagreement (ml_score >= 0.975):

- Heuristic + ML: 1,726 events
- Heuristic only: 311 events
- ML only: 2,005 events
- Neither strong: 145,197 events

Example interpretation:

- Heuristic + ML events are usually the strongest review candidates.
- Heuristic only events may be explainable rule hits that are not unusual in the wider feature space.
- ML only events may contain multivariate anomalies, but they need careful review because unusual legitimate behaviour can also be anomalous.

7. Operational Actions

Bot Hunter separates prediction from action by assigning operational tiers:

- suppress: 1,863 events
- quarantine: 1,868 events
- monitor: 145,508 events

Recommended use:

- suppress: high-confidence traffic that can be excluded from billing or downstream metrics after policy approval.
- quarantine: suspicious traffic that should be reviewed, sampled, or delayed before suppression.
- monitor: traffic not selected for bot action, retained for trend analysis and future labels.

The binary is_bot field is the compatibility output. The tier is the safer business-control layer.

8. Probability Perspective

The estimated probability that a flagged event is fraudulent is 74%. This is an operational estimate based on signal agreement, not a calibrated probability.

The estimate is stronger when independent signals agree. For example, a click with repeated query/domain replay, same-second burst evidence, and an upper-tail ML score is more likely to be fraudulent than a click selected only because one feature is unusual.

The estimate is weaker for isolated ML-only events, because unsupervised anomaly detection can also surface legitimate edge cases such as unusual campaigns, regional spikes, testing traffic, or rare but valid user behaviour.

9. Generalisation And Trade-Offs

The approach should generalise when bot traffic is repetitive, mechanically timed, or concentrated in narrow device and query patterns. It may miss bots that deliberately randomise timing, distribute across many query/domain pairs, or imitate human behaviour closely.

The main trade-off is explainability versus coverage. Rules are easier to defend but can miss novel patterns. The anomaly model broadens coverage but needs careful interpretation because it finds unusual events, not necessarily fraudulent events.

Material changes in geography, campaign volume, inventory, browser mix, or feature extraction should trigger fresh threshold review.

10. Known Limitations

- There are no ground-truth labels, so measured precision, recall, and calibration cannot be reported.
- Legitimate campaigns can create high repetition, dense clusters, or regional concentration.
- The ct rule is supporting evidence only; it should not be used as a country-level blocking rule.
- The regular inter-arrival rule uses pseudo-session groups because there is no explicit user or session identifier.
- The anomaly score is relative to the current batch and should be monitored for drift across future runs.

11. Future Work

The next useful improvements are:

- labelled validation from manual review, chargebacks, or confirmed invalid traffic feedback
- campaign-level or inventory-level normalisation
- stronger browser or user-agent fingerprinting if those fields become available
- historical drift monitoring for score distributions and flagged rates
- calibrated probabilities once trusted labels exist
- a feedback loop from reviewed suppress and quarantine decisions

12. Submission Summary

The repository includes submission.tsv with event_id, is_bot, and operational_tier.

This run selected 3,731 of 149,239 events as likely bots.
That is 2.50% of the run.

Operational split:

- suppress: 1,863
- quarantine: 1,868
- monitor: 145,508

Use the report and dashboard as review aids. They explain why traffic was selected and where the evidence is strongest, but they do not replace labelled validation or policy approval.